

Pros and Cons

Sequential models are very basic. They are simple to use since they allow you to create models layer-by-layer by stacking one layer on top of the other.

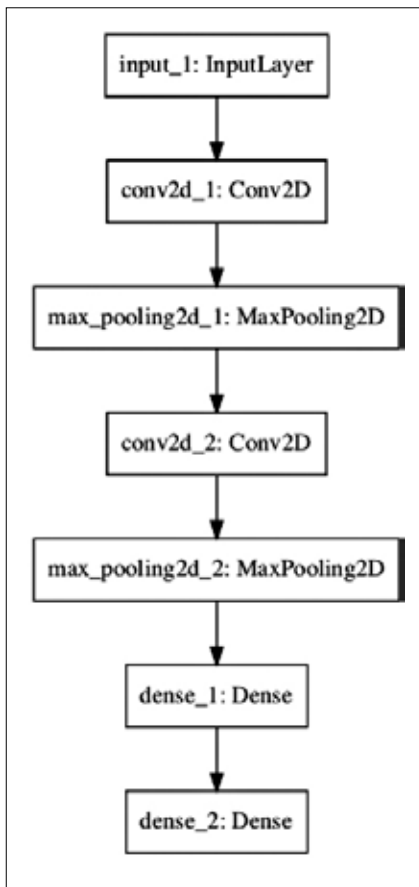
However, they have certain cons. Although there are various layers like Concatenate, Merge, Add and many more that allow for a combination of different types of models. However, it is not very flexible to make entirely different types of models, especially those with multi-input, multi-output or shared-layers like ResNet models.

Sequential model methods

Various methods or functions are used to perform various tasks in machine learning like compiling, training, evaluating and predicting the results from a model. Some of the commonly used

sequential model methods are `compile`, `fit`, `evaluate`, `predict`, `test_on_batch`, `predict_on_batch`, `fit_generator`, `evaluate_generator`, `predict_generator` and `get_layer`. All of them are discussed below along with their functions, arguments and their return types.

1. **Compile:** It is used to configure the sequentially build a model for training on the dataset. It takes various arguments such as:
 - **Optimiser:** This tells the compile method the type of optimiser to be used in backpropagation algorithm of the neural network. Various optimisers are available for use such as - Stochastic gradient descent



An example of a sequential model